

1. What index value does the third element of an array have?

The third index value would be 2.

2. Write the declaration for an array named quantities that stores 20 integers
`int [ ] quantities = new int [20];`

3. Write a declaration for an array named heights storing the numbers 1.65, 2.15, and 4.95.

`double [ ] heights = {1.65, 2.15, 4.95};`

4. Write a for-each statement that displays the integer values stored in an array named grades.

```
for (int grade : grades) {  
    System.out.println(grade);  
}
```

6. How does passing an entire array to a method differ from passing a single element of the array?

Passing the array allows the method to work with all elements and modify it. Passing a single element only gives the method a copy of that one value.

7. Why are offset array indexes required in some cases?

Because array indexes always start at zero, but the data we're working with in real life doesn't always start at zero, so we need to use the offset array indexes to get the actual value we are looking for.

8. What output is displayed by the statements below?

```
String name = "Elaine"; System.out.println(name.charAt(3));
```

The output would be the letter 'i'. Because `charAt(3)` will get the character at index 3 and it starts at 0.

10. Give an example of when a dynamic array might be a better structure choice over an array.

When you don't know the elements, or it changes during the program.

12. How can the values of wrapper class objects be compared?

It can be used by using `equals()`.